

Module-01**RBT Level-L1, L2****Introduction to Data Visualization & Power BI-**

Introduction to ETL, Data Warehouse: OLTP (Online Transaction Processing) & OLAP (Online Analytical Processing), Introduction to power BI and cloud architecture, Power BI Tools and Products, Building Block in Power BI, Software Development Life Cycle (SDLC)

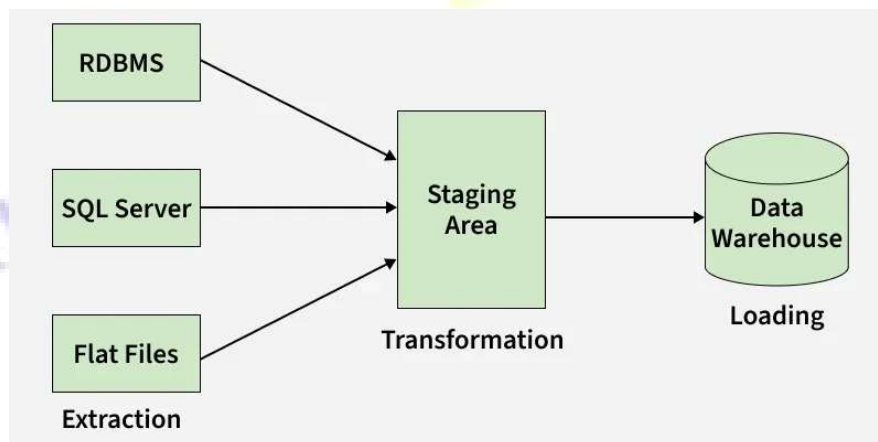
Introduction to ETL

ETL (**Extract, Transform, Load**) is a key process in data warehousing that prepares data for analysis. It involves:

- ❖ **Extracting** data from multiple sources
- ❖ **Transforming** it into a consistent format
- ❖ **Loading** it into a central data warehouse or data lake

ETL Process

The ETL process, which stands for Extract, Transform and Load, is a critical methodology used to prepare data for storage, analysis and reporting in a data warehouse. It involves three distinct stages that help to streamline raw data from multiple sources into a clean, structured and usable form.

**1. Extraction**

The Extract phase is the first step in the ETL process, where raw data is collected from various data sources. These sources can be diverse, ranging from structured sources like databases (SQL, NoSQL), to semi-structured data like JSON, XML or unstructured data such as emails or flat files. The main goal of extraction is to gather data without altering its format, enabling it to be further processed in the next stage.

Types of data sources can include:

- ❖ **Structured:** SQL databases, ERPs, CRMs
- ❖ **Semi-structured:** JSON, XML
- ❖ **Unstructured:** Emails, web pages, flat files

2. Transformation

The Transform phase is where the magic happens. Data extracted in the previous phase is often raw and inconsistent. During transformation, the data is cleaned, aggregated and formatted according to business rules. This is a crucial step because it ensures that the data meets the quality standards required for accurate analysis.

Common transformations include:

- ❖ **Data Filtering:** Removing irrelevant or incorrect data.
- ❖ **Data Sorting:** Organizing data into a required order for easier analysis.
- ❖ **Data Aggregating:** Summarizing data to provide meaningful insights (e.g., averaging sales data).

3. Loading

Once data has been cleaned and transformed, it is ready for the final step: Loading. This phase involves transferring the transformed data into a data warehouse, data lake or another target system for storage. Depending on the use case, there are two types of loading methods:

- ❖ **Full Load:** All data is loaded into the target system, often used during the initial population of the warehouse.
- ❖ **Incremental Load:** Only new or updated data is loaded, making this method more efficient for ongoing data updates.

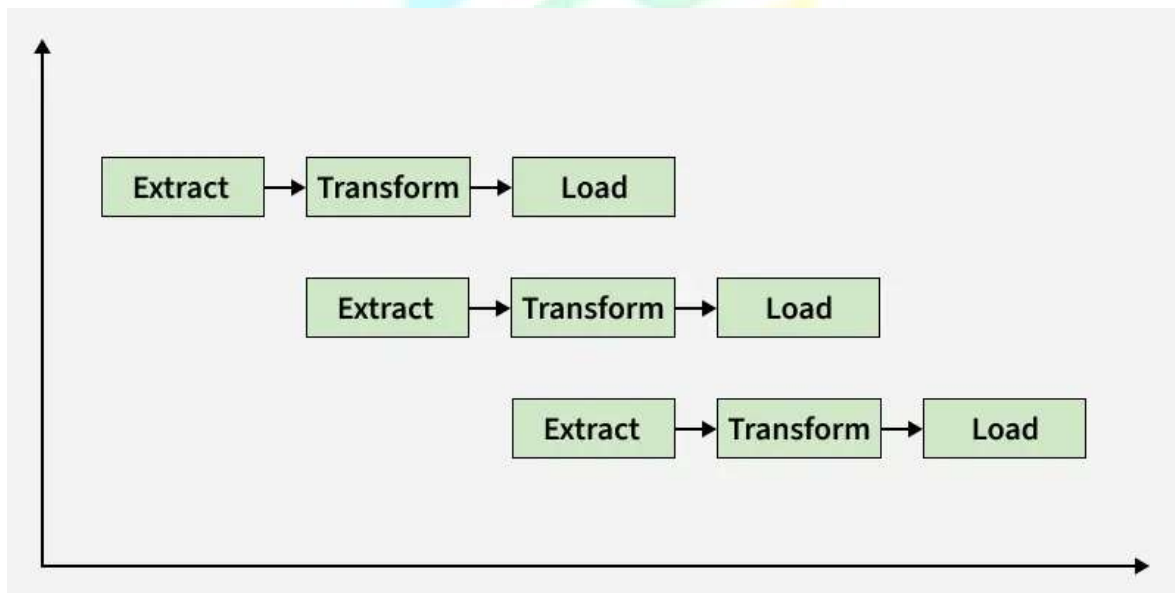
Difference between Datawarehouse and data lake

Basis of Comparison	Data Lake	Vs	Data Warehouse
Data Structure	Raw		Structured
Purpose of Data	Still to be determined		Currently in use
Users	Data Scientists		Business Professionals
Accessibility	Highly accessible and updated quickly		Changes are more difficult and expensive to implement

Pipelining in ETL Process

Pipelining in the ETL process involves processing data in overlapping stages to enhance efficiency. Instead of completing each step sequentially, data is extracted, transformed and loaded concurrently. As soon as data is extracted, it is transformed and while transformed data is being loaded into the warehouse, new data can continue being extracted and processed.

- ❖ This parallel execution reduces downtime, speeds up the overall process and improves system resource utilization, making the ETL pipeline faster and more scalable.
- ❖ In short, the ETL process involves extracting raw data from various sources, transforming it into a clean format and loading it into a target system for analysis.



Importance of ETL

- ❖ **Data Integration:** ETL combines data from various sources, including structured and unstructured formats, ensuring seamless integration for a unified view.
- ❖ **Data Quality:** By transforming raw data, ETL cleanses and standardizes it, improving data accuracy and consistency for more reliable insights.
- ❖ **Essential for Data Warehousing:** ETL prepares data for storage in data warehouses, making it accessible for analysis and reporting by aligning it with the target system's requirements.

- ❖ **Enhanced Decision-Making:** ETL helps businesses derive actionable insights, enabling better forecasting, resource allocation and strategic planning.
- ❖ **Operational Efficiency:** Automating the data pipeline through ETL speeds up data processing, allowing organizations to make real-time decisions based on the most current data.

Challenges in ETL Process

The ETL process, while essential for data integration, comes with its own set of challenges that can hinder efficiency and accuracy. These challenges, if not addressed properly, can impact the overall performance and reliability of data systems.

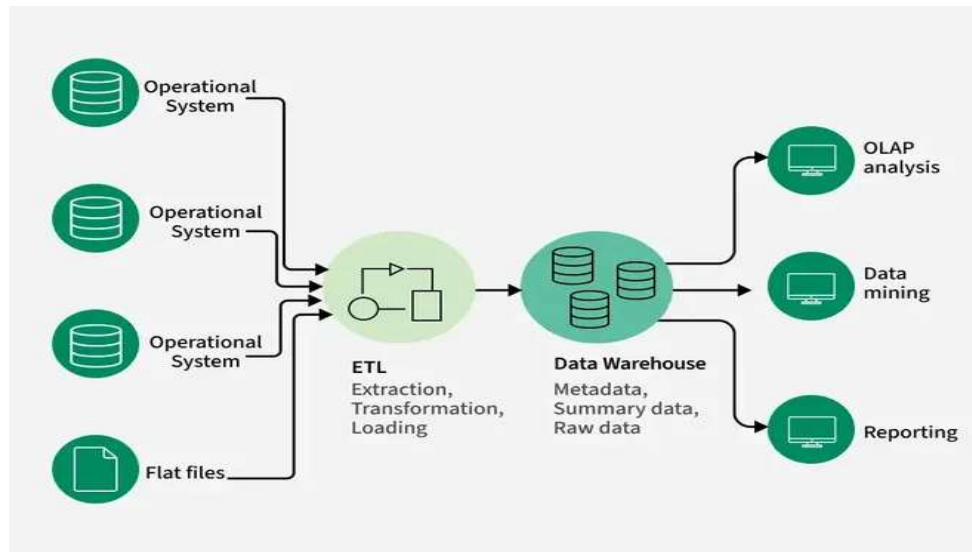
- ❖ **Data Quality Issues:** Inconsistent, incomplete or duplicate data from multiple sources can impact transformation and loading, leading to inaccurate insights.
- ❖ **Performance Bottlenecks:** Large datasets can slow down or cause ETL processes to fail, particularly during complex transformations like cleansing and aggregation.
- ❖ **Scalability Issues:** Legacy ETL systems may struggle to scale with growing data volumes, diverse sources and more complex transformations.

Data Warehouse: OLTP(Online Transaction Processing) & OLAP(Online Analytical Processing)

Data Warehousing

Data warehousing is the process of collecting, integrating, storing and managing data from multiple sources in a central repository. It enables organizations to organize large volumes of current and historical data for efficient querying, analysis and reporting.

Data warehousing Architecture



- ❖ **Data Sources:** These are the various operational systems, databases and external data feeds that provide raw data to be stored in the warehouse.
- ❖ **ETL (Extract, Transform, Load) Process:** The ETL process is responsible for extracting data from different sources, transforming it into a suitable format and loading it into the data warehouse.
- ❖ **Data Warehouse Database:** This is the central repository where cleaned and transformed data is stored. It is typically organized in a multidimensional format for efficient querying and reporting.
- ❖ **Metadata:** Metadata describes the structure, source and usage of data within the warehouse, making it easier for users and systems to understand and work with the data.
- ❖ **Data Marts:** These are smaller, more focused data repositories derived from the data warehouse, designed to meet the needs of specific business departments or functions.
- ❖ **OLAP (Online Analytical Processing) Tools:** OLAP tools allow users to analyze data in multiple dimensions, providing deeper insights and supporting complex analytical queries.
- ❖ **End-User Access Tools:** These are reporting and analysis tools, such as dashboards or Business Intelligence (BI) tools, that enable business users to query the data warehouse and generate reports.

Types of Data Warehouses

- ❖ **Enterprise Data Warehouse (EDW):** A centralized warehouse that stores data from across the organization for analysis and reporting.
- ❖ **Operational Data Store (ODS):** Stores real-time operational data used for day-to-day operations, not for deep analytics.

- ❖ **Data Mart:** A subset of a data warehouse, focusing on a specific business area or department.
- ❖ **Cloud Data Warehouse:** A data warehouse hosted in the cloud, offering scalability and flexibility.
- ❖ **Big Data Warehouse:** Designed to store vast amounts of unstructured and structured data for big data analysis.
- ❖ **Virtual Data Warehouse:** Provides access to data from multiple sources without physically storing it.
- ❖ **Hybrid Data Warehouse:** Combines on-premises and cloud-based storage to offer flexibility.
- ❖ **Real-time Data Warehouse:** Designed to handle real-time data streaming and analysis for immediate insights.

OLAP (Online Analytical Processing) and OLTP (Online Transaction Processing) are both integral parts of data management, but they have different functionalities.

- ❖ OLTP focuses on handling large numbers of transactional operations in real time, ensuring data consistency and reliability for daily business operations.
- ❖ OLAP is designed for complex queries and data analysis, enabling businesses to derive insights from vast datasets through multidimensional analysis.

Online Analytical Processing (OLAP)

Online Analytical Processing (OLAP) refers to software tools used for the analysis of data in business decision-making processes. OLAP systems generally allow users to extract and view data from various perspectives, many times they do this in a multidimensional format which is necessary for understanding complex interrelations in the data.

Benefits of OLAP Services

- ❖ Helps in keeping consistency and performing calculation on data.
- ❖ Can store planning, analysis and budgeting for business analytics within one platform.
- ❖ Efficiently handle large volumes of data, making them suitable for enterprise-level business applications.
- ❖ Assist in applying security restrictions for data protection.
- ❖ Provide a multidimensional view of data, which helps in applying operations on data in various ways.

Drawbacks of OLAP Services

- ❖ Requires professionals to handle the data because of its complex modeling procedure.
- ❖ Expensive to implement and maintain in cases when datasets are large.
- ❖ Data analysis occurs only after extraction and transformation, leading to system delays.
- ❖ Not efficient for decision-making, as it is updated on a periodic basis.

Online Transaction Processing (OLTP)

Online Transaction Processing, commonly known as OLTP, is a data processing approach emphasizing real-time execution of transactions. The majority of OLTP systems are meant to manage numerous short atomic operations that keep databases in line.

- ❖ To maintain transaction integrity and reliability, these systems support ACID (Atomicity, Consistency, Isolation, Durability) properties.
- ❖ It is through this that numerous unavoidable applications run their critical courses like online banking, reservation systems etc.

Benefits of OLTP Services

- ❖ Allow users to quickly read, write and delete data operations.
- ❖ Support an increase in users and transactions for real-time data access.
- ❖ Provide better data protection through multiple security features.
- ❖ Aid in decision-making with accurate, up-to-date data.
- ❖ Ensure data integrity, consistency and high availability.

Drawbacks of OLTP Services

- ❖ Limited analysis capability, not suited for complex analysis or reporting.
- ❖ High maintenance costs due to frequent updates, backups and recovery.
- ❖ Susceptible to disruption during hardware failures, impacting online transactions.
- ❖ Prone to issues like duplicate or inconsistent data.

Difference Between OLAP and OLTP

Category	OLAP (Online Analytical Processing)	OLTP (Online Transaction Processing)
Data Source	Historical data from multiple databases.	Current operational data.
Purpose	Used for analysis and decision-making.	Used for day-to-day transactions.
Method Used	Uses a data warehouse.	Uses a standard DBMS.

Normalization	Tables are not normalized.	Tables are normalized (3NF).
Query Type	Complex, read-heavy queries (slow).	Simple, read/write queries (fast).
Data Volume	Large (TB-PB).	Small (MB-GB).
Update Frequency	Updated periodically in batches.	Updated frequently by users.
Backup & Recovery	Periodic backup.	Continuous and rigorous backup.
Users	Used by analysts, managers and executives.	Used by clerks and operational staff.
Focus	Subject-oriented (analysis-focused).	Application-oriented (operation-focused).

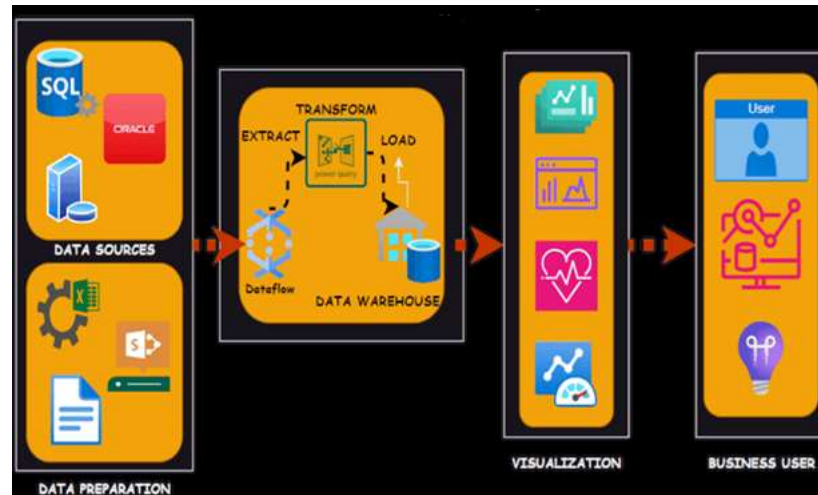
Introduction to power BI and cloud architecture

Power BI is a Microsoft-powered business intelligence tool that helps transform raw data into interactive dashboards and actionable insights. It allow users to connect to various data sources, clean and shape data and visualize it using charts, graphs and reports all with minimal coding.

What is Power BI Used For?

- ❖ Bring in data from different places like Excel files, SQL databases, CSVs, JSON files and even websites.
- ❖ Clean and fix your data easily without writing code.
- ❖ Create visuals like bar charts, line graphs, pie charts and dashboards to help you see patterns and trends.
- ❖ Analyze your data using filters and slicers, so you can focus on specific details like sales by region or product performance over time.
- ❖ Share your dashboards with your team or clients so everyone stays informed and can explore the data on their own.
- ❖ Set up automatic updates so your reports always show the latest information without you having to do anything.

Power BI & Cloud architecture



The PowerBI architecture is divided into three stages. The data is handled in the first two phases using the ETL (extract, transform, and load) method.

Data Processing:

- ❖ The data that comes from a **variety of sources**, such as your **separate database, servers, etc.**
- ❖ The data is then incorporated into a **standard format** and **stored in a shared area** known as the **staging area**.

Data Processing:

- ❖ The combined data is **still not ready for visualization** because the data must be **processed before it can be displayed**.
- ❖ This data has been **pre-processed**, **Missing or redundant values**, will be deleted from the data sets.
- ❖ Following that, **business rules** are applied to the data and transformed into **presentable data**.
- ❖ The data will then be **imported into the data warehouse**.

Data presentation:

- ❖ Once the data has been **loaded and processed**, it may be more effectively represented using the many **visualization tools provided by Power BI**.

Power BI Products



Data Sources

- ❖ Power BI connects to a vast range of data sources, from cloud platforms like Azure, AWS, and Salesforce to on-premises databases such as SQL Server and Oracle. It even integrates with simple Excel spreadsheets. Using dedicated connectors, Power BI ensures seamless data flow, translating information from diverse systems into a common language for accurate, up-to-date insights.

Power Query (ETL)

- ❖ Power Query is Power BI's ETL engine (Extract, Transform, Load). It enables users to pull raw data, clean it, and shape it into a ready-to-use format—all without advanced coding. This process ensures that data is consistent, tailored, and optimized for analysis, empowering organizations to uncover actionable insights.

Data Model

- ❖ The data model is the structural backbone of Power BI, linking tables to enable complex analysis. By using a star schema, Power BI supports multi-dimensional queries, improving both the accuracy of insights and the efficiency of report creation.

Power BI Desktop

- ❖ This is the primary tool for building data models, creating reports, and designing visualizations. Its intuitive interface combines ease of use with robust functionality, allowing users to create meaningful insights that drive decisions.

Power BI Service

- ❖ A cloud-based platform for sharing and collaborating on reports and dashboards. It supports real-time data refreshes, making insights accessible to decision-makers anytime, anywhere. Power BI Service is key to fostering collaboration and ensuring timely business decisions.

Power BI Gateway

- ❖ Power BI Gateway bridges on-premises data sources with the Power BI cloud. This secure connection allows organizations to leverage sensitive, locally-stored data without compromising accessibility or security.

Power BI Report Server

- ❖ Ideal for companies needing on-premises BI solutions, Power BI Report Server offers the same flexibility and capabilities as the cloud version while ensuring that data and reports remain within organizational boundaries.

Power BI Mobile

- ❖ Power BI Mobile extends accessibility to smartphones and tablets, allowing executives and managers to monitor KPIs on the go. With real-time dashboards optimized for mobile, decisions can be made quickly, even away from the office.

DAX (Data Analysis Expressions)

- ❖ DAX is the formula language behind Power BI's custom calculations. It empowers users to create advanced metrics and tailor data insights to their specific needs, enabling deeper analysis.

Row-Level Security (RLS)

- ❖ RLS restricts data access based on user roles, ensuring that employees only see information relevant to their responsibilities. This feature enhances security and compliance while maintaining data privacy.

Power BI API

- ❖ The API allows developers to customize Power BI, automate workflows, and integrate reports into other systems. It streamlines processes, ensuring that dashboards and data remain current and accessible.

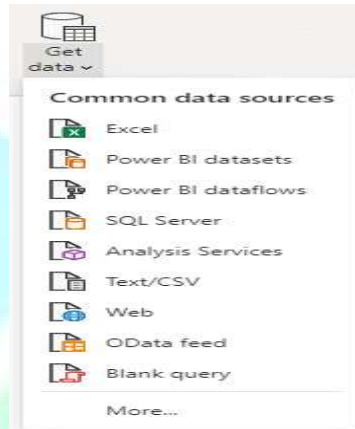
Power BI Embedded

- ❖ Power BI Embedded enables organizations to integrate reporting and visualization tools directly into their applications. This feature enhances user experience by providing customers and stakeholders with seamless access to data insights.

Building Block in Power BI.

There are five Power BI building blocks which are:

1. DataSets



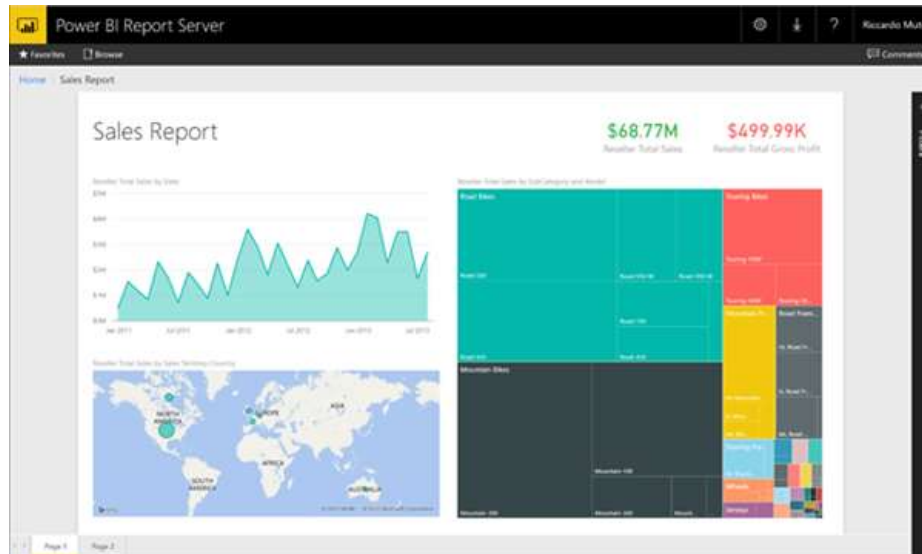
- ❖ Dataflow's source data or the collection of data that are imported are called Datasets. Each of these datasets is connected to Power BI to make them be stored in one single place.

2. Visualization



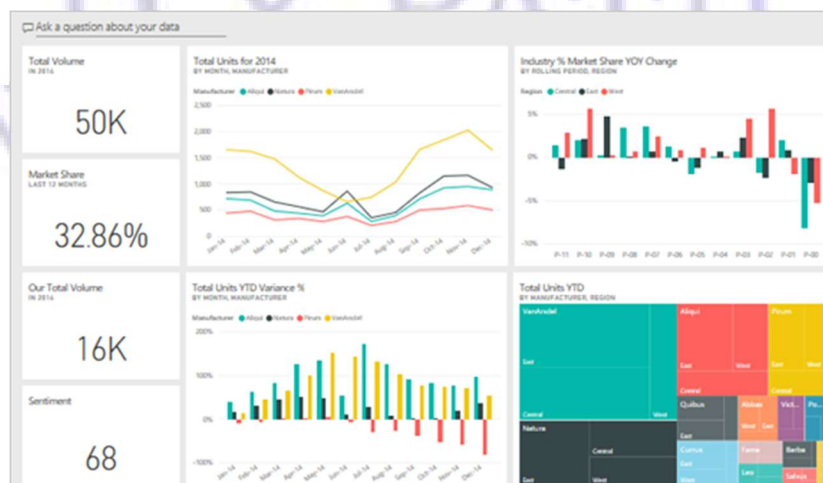
- ❖ When the data are represented in the form of charts/graphs/diagrams/ any other form of visuals, it is called visualization. These help in reforming the data for a better understanding.

2. Reports



- ❖ The visual representation of any collected data is called Reports and they include charts, maps, graphs, and even tree maps. These reports can be created in Power BI and can be shared through dashboards, with one's friends. Another way to generate reports is when the user connects to various datasets from their SaaS applications / Excel workbooks. Report making includes 2 modes, called Reading View and Editing View.

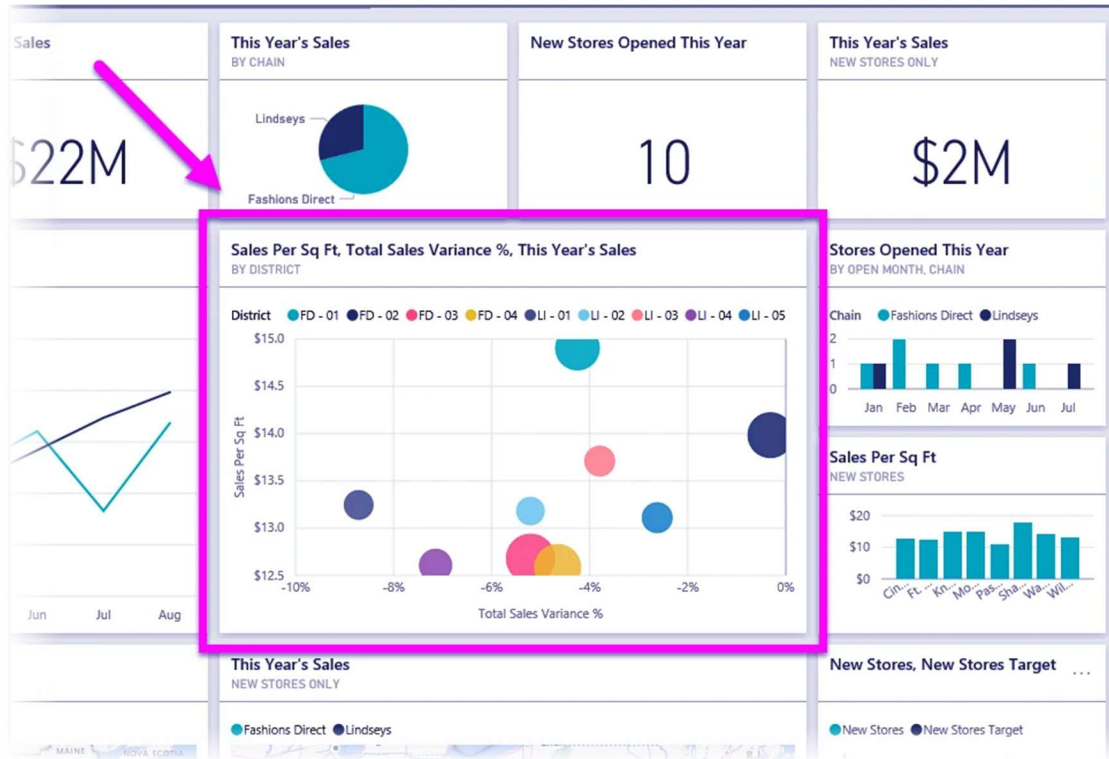
3. Dashboards



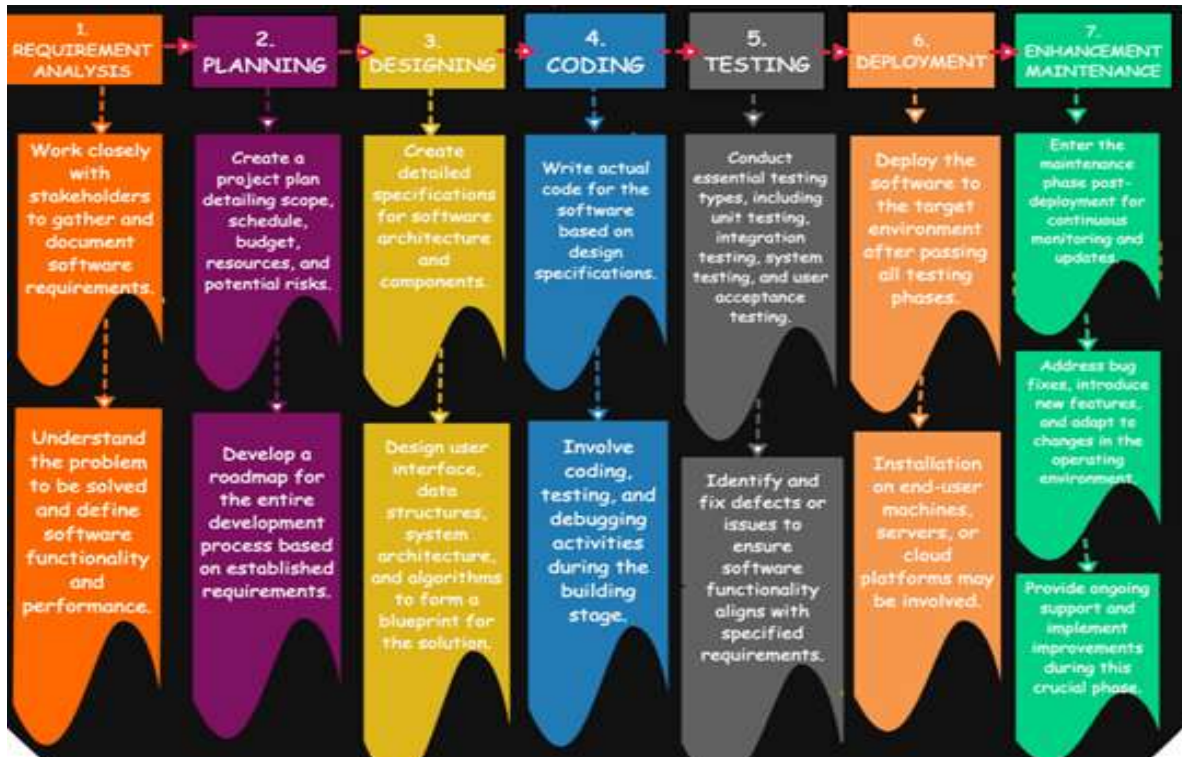
- ❖ A single sheet that consists of numerous widgets and tiles is called Dashboards. These are either created by the user himself or are borrowed from the one created and shared by their colleague in the Power BI service. In short, it is a collection of various data and reports.

5. Tiles

- ❖ Each and every visual data stored in the dashboards are called a Tile. In other words, the Dashboard is a collection of Tiles. In Power BI, a tile is a solitary representation found in a report or on a dashboard. It's the rectangular box that contains every individual visual.



SOFTWARE DEVELOPMENT LIFE CYCLE (SDLC)



1. Requirement Analysis:

The Requirement Analysis phase in the Software Development Life Cycle is crucial and is often viewed as the cornerstone of a successful project. This phase involves meticulously gathering and scrutinizing the requirements from the client or end users. The key here is to understand what the users expect from the software in terms of functionality, performance, and usability.

2. Design

The Design phase in the Software Development Life Cycle (SDLC) is a critical stage where the requirements gathered and analyzed in the previous phase are transformed into a blueprint for constructing the software. This phase involves two key aspects:

a. Architectural Design (High-Level Design):

In this step, the overall software architecture is defined.

It involves outlining the software's major components, their interactions, and how they fit into the larger system. This high-level design provides a bird's-eye view of the solution, including database schemas, software layers and modules, technology stack (like programming languages, frameworks, and tools), and the overall system architecture (like client-server, microservices, etc.).

b. Detailed Design (Low-Level Design):

In software, this step involves deciding the main parts of the software and how they will interact. During the Detailed Design (Low-Level Design) once the overall layout is set, more detailed plans are made

3. Coding:

The Implementation or Coding phase in the Software Development Life Cycle (SDLC) is where the actual construction of the software takes place. The Implementation or Coding phase in software development can be compared to the actual building process in construction.

a. Writing Code:

- ❖ This is like the actual building work. Developers start writing code, each line of code is a building block for the software. They use programming languages (like Java, Python, etc.).

b. Unit Testing:

- ❖ This means they check small parts of the code to ensure they work properly.

c. Integration:

- ❖ In software, this is about integrating different code modules and ensuring they work together smoothly.

d. Review and Revision:

- ❖ The software code is reviewed by other developers who suggest improvements or fix any issues. In this phase, the software goes from being a design and a plan

4. Testing:

The Testing phase the software is thoroughly checked to make sure it works correctly.

a. Identifying and Fixing Bugs:

- ❖ In software, this is about finding and fixing bugs, which are errors or problems in the code.

b. Various Types of Testing:

In software testing, different types focus on different parts:

- ❖ Unit Testing is like checking individual elements (like a single window or a light fixture).
- ❖ Integration Testing is making sure different parts (like the electrical system and the plumbing) work well together.
- ❖ System Testing is examining the whole house as a complete unit.

- ❖ Acceptance Testing is where the future homeowner checks if the house meets their expectations.

c. Ensuring Quality and Compliance:

- ❖ The aim is to ensure that everything in the software is working as it should, similar to how an inspection ensures that a house is safe, functional, and ready for living.

d. Regression Testing:

- ❖ In software, regression testing ensures that new changes don't introduce new problems.

5. Deployment:

Software development is planning, building, and preparation, Here's the breakdown:

a. Releasing the Software:

- ❖ In software terms, it means making the software available for its intended users. This could involve installing the software on user devices, releasing it on a website for download, or deploying it on servers for users to access online.

b. Different Deployment Strategies:

- ❖ In software where the software is rolled out in stages, like a beta release for a limited audience before a full-scale launch.

c. Ensuring Proper Integration:

- ❖ The software integrates well with the users' existing systems.

d. Training and Support:

- ❖ New users of the software might need training or guidance. Providing user manuals, FAQs, or customer support is part of this phase.

e. Monitoring and Updating:

- ❖ After deploying software, it's important to monitor its performance, fix any issues, and make updates to improve functionality or security.

6. Maintenance:

This phase is crucial for ensuring the software continues to function correctly and efficiently over time. software needs regular maintenance after it's been deployed.

a. Bug Fixing:

- ❖ In software, issues, known as bugs, can appear over time. Fixing these bugs is an essential part of software maintenance.

b. Updates and Upgrades:

- ❖ software needs updates and upgrades to add new features, improve performance, or enhance security.

c. Performance Optimization:

- ❖ Software needs regular checks to ensure it's performing well. This might involve making it faster, more efficient, or easier to use.

d. Adapting to Changing Environments:

- ❖ Software might need changes to work well with new operating systems, devices, or other software.

e. User Support and Documentation:

- ❖ Software users might need support for using the software or solving issues. Keeping user guides and help documents up to date is also part of maintenance.

In summary, the Maintenance phase is about ensuring the software continues to function correctly, stay relevant, and meet user needs, just like regular maintenance and updates keep a house safe, functional, and up-to-date.

S.No	Questions
01	Define ETL? With neat block diagram, explain ETL.
02	List difference between Datawarehouse and data lake.
03	Explain the Pipelining in ETL Process.
04	Explain Importance and Challenges in ETL Process?
05	With neat block diagram, explain Data warehousing Architecture.
06	List and explain different types of Data Warehouses
07	Define Online Analytical Processing (OLAP)? Mention Benefits & Drawbacks of OLAP Services.
08	Define Online Transaction Processing (OLTP)? Mention Benefits & Drawbacks of OLTP Services.
09	Compare between OLAP and OLTP
10	Define Power BI? With neat block diagram, explain PowerBI architecture.
11	List and explain different types of Power BI Products
12	Explain Building Block in Power BI.
13	With neat block diagram, explain Software Development Life Cycle (SDLC)